Grade 9 Practice

Standard A.SSE.1

Name: _____ Date: _____

Computation

1. In 3x + 7, what is the coefficient of x? _____

3

2. In $2x^2 - 5$, what is the constant term? _____

-5

3. How many terms are in 4x - 9? _____

2

4. In 5(x + 2), what is the factor outside the parentheses? _____

5

Word problems

5. The cost is C = 12h + 30, where h is hours. What does 12 represent?

Show your work and answer.

Answer: the rate per hour

6. The value is V = 500 - 25t, where t is days. What does 25 represent?

Show your work and answer.

Answer: the rate of decrease per day

Grade 9 Practice

Standard A.SSE.2

Name: _____ Date: _____

Computation

1. Rewrite $x^2 - 9$ as a product of two factors. _____

(x − 3)(x + 3)

2. Factor $x^2 + 5x + 6$. _____

(x + 2)(x + 3)

3. Factor $4x^2 - 1$. _____

(2x − 1)(2x + 1)

4. Rewrite $a^2 + 2ab + b^2$ as a square of a binomial. _____

$(a + b)^2$

Word problems

5. A square patio has area $x^2 + 6x + 9$. What is the side length?

Show your work and answer.

Answer: **x + 3**

6. Factor $9 - x^2$.

Show your work and answer.

Answer: **(3 − x)(3 + x)**

Grade 9 Practice

Standard A.APR.1

Name: _____ Date: _____

Computation

1. Simplify $(x + 3) + (2x - 1)$. _____

3x + 2

2. Multiply $(x + 2)(x + 5)$. _____

$x^2 + 7x + 10$

3. Simplify $(3x - 4) - (x + 6)$. _____

2x − 10

4. Multiply $2x(x - 3)$. _____

$2x^2 - 6x$

Word problems

5. A rectangle has sides $x + 2$ and $x + 4$. What is its area?

Show your work and answer.

Answer: **$x^2 + 6x + 8$**

6. Add $(2x^2 + 3)$ and $(x^2 - x)$. What is the sum?

Show your work and answer.

Answer: **$3x^2 - x + 3$**

Grade 9 Practice

Standard A.CED.1

Name: _____ Date: _____

Computation

1. Write an equation: 5 more than 3 times n is 20. _____

3n + 5 = 20

2. Write an equation: twice x minus 7 equals 11. _____

2x − 7 = 11

3. Write an inequality: y is at least 4. _____

y ≥ 4

4. Write an equation: x divided by 4 equals 6. _____

x/4 = 6

Word problems

5. A phone plan costs $15 plus $0.10 per text. Your bill is $25. Write an equation using t for texts.

Show your work and answer.

Answer: **15 + 0.10t = 25**

6. A team scored 3 times its points p plus 4 points, for a total of 31. Write an equation.

Show your work and answer.

Answer: **3p + 4 = 31**

Grade 9 Practice

Standard A.CED.2

Name: _____ Date: _____

Computation

1. In the line y = 2x + 1, what is the y-intercept? _____

1

2. In the line y = -x + 4, what is the slope? _____

-1

3. Is the point (2, 5) on the line y = 2x + 1? Write yes or no.

yes

4. Is the point (0, 3) on the line y = x - 3? Write yes or no.

no

Word problems

5. A cost is y = 5x + 10, where x is hours. What does the 10 represent?

Show your work and answer.

Answer: a fixed cost of 10

6. A car's distance is y = 60x, where x is hours. What does the slope represent?

Show your work and answer.

Answer: 60 miles per hour

Grade 9 Practice

Standard A.CED.3

Name: _____ Date: _____

Computation

1. Solve the system: x + y = 10 and x - y = 4. Write (x, y).

(7, 3)

2. Solve the system: y = 2x and y = x + 3. Write (x, y).

(3, 6)

3. Pens cost p and pencils cost c. 3p + 2c = 11 and p + 2c = 7. What is the cost of one pen? _____

2

4. Adult tickets a and child tickets c: a + c = 50 and a - c = 10. What is a? _____

30

Word problems

5. Tickets: a + c = 100 and 8a + 5c = 620, where a is adult and c is child tickets. How many of each were sold?

Show your work and answer.

Answer: 40 adult, 60 child

6. Nickels n and dimes d: n + d = 12 and 5n + 10d = 85. How many nickels and how many dimes?

Show your work and answer.

Answer: 7 nickels, 5 dimes

Grade 9 Practice

Standard A.CED.4

Name: _____ Date: _____

Computation

1. Solve y = mx + b for b. b = _____

y - mx

2. Solve 2x + y = 6 for y. y = _____

6 - 2x

3. Solve A = lw for w. w = _____

A/l

4. Solve C = 2πr for r. r = _____

C/(2π)

Word problems

5. Distance d = rt. Solve for t.

Show your work and answer.

Answer: t = d/r

6. Perimeter P = 2l + 2w. Solve for w.

Show your work and answer.

Answer: w = (P - 2l)/2

Grade 9 Practice

Standard A.REI.1

Name: _____ Date: _____

Computation

1. Solve x/2 = 5. x = _____

10

2. Solve 4x - 8 = 0. x = _____

2

3. Solve 5 = x - 3. x = _____

8

4. Solve 3x = -9. x = _____

-3

Word problems

5. Sam says x = 2 solves 3x + 1 = 7. Is he right? Write yes or no.

Show your work and answer.

Answer: yes

6. Mia solves 2x + 3 = 11 by subtracting 3 and then dividing by 2. Does she get x = 4? Write yes or no.

Show your work and answer.

Answer: yes

Grade 9 Practice

Standard A.REI.3

Name: _____ Date: _____

Computation

1. Solve $x + 3 > 7$. $x >$ _____ **4**

2. Solve $2x \leq 10$. $x \leq$ _____ **5**

3. Solve $-3x < 9$. $x >$ _____ **-3**

4. Solve $4x - 5 \geq 11$. $x \geq$ _____ **4**

Word problems

5. Tickets cost $8 each. You can spend at most $64. Write an inequality for t tickets and solve it.

Show your work and answer.

Answer: **$t \leq 8$**

6. A game needs at least 30 points. You earn 4 points per round. What is the fewest whole number of rounds needed?

Show your work and answer.

Answer: **8 rounds**

Grade 9 Practice

Standard A.REI.4

Name: _____ Date: _____

Computation

1. Solve $x^2 = 16$. x = _____

±4

2. Solve $x^2 - 9 = 0$. x = _____

±3

3. Solve $(x - 2)^2 = 25$. x = _____

7 or -3

4. Solve $x^2 = 0$. x = _____

0

Word problems

5. A square has area 49 square units. What is its side length?

Show your work and answer.

Answer: 7 units

6. A square has area 81 square units. What is its side length?

Show your work and answer.

Answer: 9 units

Grade 9 Practice

Standard A.REI.5

Name: _____ Date: _____

Computation

1. Does multiplying x + y = 4 by 2 give 2x + 2y = 8? Write yes or no.

yes

2. Is 2x = 6 equivalent to x = 3? Write yes or no.

yes

3. Multiply x + y = 4 by 2. Write the new equation.

2x + 2y = 8

4. Is 3x = 6 equivalent to 6x = 12? Write yes or no.

yes

Word problems

5. Solve 2x - 4 = 6 and 4x - 8 = 12. Do they have the same solution? Write yes or no.

Show your work and answer.

Answer: yes

6. If you multiply both sides of an equation by 0, do the solutions stay the same? Write yes or no.

Show your work and answer.

Answer: no

Grade 9 Practice

Standard A.REI.6

Name: _____ Date: _____

Computation

1. Solve the system: y = x + 2 and y = -x. Write (x, y).

(-1, 1)

2. Solve the system: y = 2x + 1 and y = 2x + 4. Write yes if there is a solution or no if there is none.

no

3. Solve the system: x + y = 6 and x - y = 2. Write (x, y).

(4, 2)

4. Solve the system: y = 3x and y = x + 4. Write (x, y).

(2, 6)

Word problems

5. Plan A costs y = 5x + 10 and plan B costs y = 3x + 30. For what x do the costs match?

Show your work and answer.

Answer: x = 10

6. Solve the system: x + y = 20 and y = 3x. Write (x, y).

Show your work and answer.

Answer: (5, 15)

Grade 9 Practice

Standard F.IF.1

Name: _____ Date: _____

Computation

1. If $f(x) = 2x + 3$, what is $f(4)$? _____ **11**

2. If $g(x) = x^2$, what is $g(-3)$? _____ **9**

3. If $h(x) = x - 5$, what is $h(10)$? _____ **5**

4. If $f(x) = 7x - 2$, what is $f(0)$? _____ **-2**

Word problems

5. A phone cost is $c(t) = 10 + 5t$, where t is months. What is $c(3)$?

Show your work and answer.

Answer: 25 dollars

6. If $f(x) = x^2 - 1$, which input gives $f(x) = 0$? Write your answer.

Show your work and answer.

Answer: x = 1 or x = -1

Grade 9 Practice

Standard F.IF.2

Name: _____ Date: _____

Computation

1. If $f(x) = 3x + 1$, find $f(2)$. _____ **7**

2. If $g(x) = x^2 + 2x$, find $g(3)$. _____ **15**

3. If $h(x) = 10 - x$, find $h(-4)$. _____ **14**

4. If $f(x) = x/2$, find $f(8)$. _____ **4**

Word problems

5. The temperature is $T(h) = 30 - 2h$, where h is hours. What is $T(5)$?

Show your work and answer.

Answer: 20 degrees

6. If $A(s) = s^2$, find $A(6)$.

Show your work and answer.

Answer: 36

Grade 9 Practice

Standard F.IF.4

Name: _____ Date: _____

Computation

1. What is the y-intercept of y = 3x - 6? _____ **-6**

2. What is the x-intercept of y = 2x - 8? _____ **4**

3. What is the slope of y = -x + 2? _____ **-1**

4. Does the graph of y = 2x increase? Write yes or no. **yes**

Word problems

5. The graph of $y = x^2$ has its lowest point at (0, 0). Is that the vertex? Write yes or no.

Show your work and answer.

Answer: yes

6. A temperature graph falls after noon. Is it increasing or decreasing after noon?

Show your work and answer.

Answer: decreasing

Grade 9 Practice

Standard F.BF.1

Name: _____ Date: _____

Computation

1. A pile starts at 10 and adds 4 each day. Write the function f(x) for x days.

f(x) = 4x + 10

2. A number is multiplied by 3. Write f(x).

f(x) = 3x

3. A number is squared and then 1 is subtracted. Write f(x).

f(x) = x² − 1

4. A number is doubled and then 5 is added. Write f(x).

f(x) = 2x + 5

Word problems

5. A phone plan costs $30 plus $5 per month. Write a function f(m) for the cost after m months.

Show your work and answer.

Answer: **f(m) = 5m + 30**

6. A bacteria count starts at 100 and doubles every hour. Write a function f(t) for t hours.

Show your work and answer.

Answer: **f(t) = 100 · 2^t**

Grade 9 Practice

Standard F.LE.1

Name: _____ Date: _____

Computation

1. A table increases by 2 at each step. Is it linear or exponential? **linear**

2. A table doubles at each step. Is it linear or exponential? **exponential**

3. A table increases by 5 at each step. Is it linear or exponential? **linear**

4. A table multiplies by 3 at each step. Is it linear or exponential? **exponential**

Word problems

5. A population grows by 10% each year. Is it linear or exponential?

Show your work and answer.

Answer: exponential

6. A salary increases by $500 each year. Is it linear or exponential?

Show your work and answer.

Answer: linear

Grade 9 Practice

Standard F.LE.2

Name: _____ Date: _____

Computation

1. A line passes through (0, 3) and (1, 5). Write the equation.

y = 2x + 3

2. An exponential function passes through (0, 2) and (1, 6). Write the equation.

y = 2 · 3^x

3. A line has slope 4 and y-intercept -1. Write the equation.

y = 4x - 1

4. An exponential function starts at 5 and doubles each step. Write the equation.

y = 5 · 2^x

Word problems

5. Savings start at $100 and grow by $20 each month. Write an equation for y after x months.

Show your work and answer.

Answer: **y = 20x + 100**

6. A value is cut in half each year, starting at 800. Write an equation for y after x years.

Show your work and answer.

Answer: **y = 800(1/2)^x**

Grade 9 Practice

Standard S.ID.1

Name: _____ Date: _____

Computation

1. Find the range of 2, 4, 6, and 8. _____ **6**

2. Find the mean of 1, 2, 3, 4, and 5. _____ **3**

3. Find the median of 1, 3, 5, and 7. _____ **4**

4. Find the mode of 2, 2, 3, and 4. _____ **2**

Word problems

5. Five scores are 80, 85, 90, 95, and 100. What is the mean?

Show your work and answer.

Answer: 90

6. A histogram has 6 values in the 10–20 bin and 4 in the 20–30 bin. How many values in all?

Show your work and answer.

Answer: 10

Grade 9 Practice

Standard S.ID.2

Name: _____ Date: _____

Computation

1. Which measure of center is resistant to outliers? Write mean or median. **median**

2. Find the mean of the data 1, 2, 3, 4, and 5. _____ **3**

3. What does standard deviation measure? Write spread or center. **spread**

4. For symmetric data, how do the mean and median compare? Write equal or not equal. **equal**

Word problems

5. Income data is skewed to the right. Which measure of center is better to use?

Show your work and answer.

Answer: **median**

6. Data are 10, 12, 14, and 16. What is the mean?

Show your work and answer.

Answer: **13**